THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 6: **Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Pressure is defined as: 1

(a) force acting per unit area **(b)** mass per unit volume
(c) force multiplied by area **(d)** area per unit force

2. The instrument used to measure atmospheric pressure is a: 1

(a) barometer **(b)** thermometer **(c)** hygrometer **(d)** speedometer

3. A cyclone has at its centre a region of: 1

(a) very low pressure **(b)** solid ice
(c) no air **(d)** very high pressure

4. The SI unit of pressure is the: 1

(a) joule **(b)** pascal **(c)** watt **(d)** newton

5. Moving air is called: 1

(a) wind **(b)** cloud **(c)** rain **(d)** fog

6. The calm centre of a cyclone is called the: 1

(a) wall **(b)** arm **(c)** eye **(d)** front

7. A force of 100 N acts on an area of 2 square metres. The pressure is: 1

(a) 98 Pa **(b)** 0.02 Pa **(c)** 50 Pa **(d)** 200 Pa

8. Winds are caused by: 1

(a) moon only
(b) differences in air pressure due to uneven heating
(c) rotation of fans
(d) rain

9. A tropical cyclone forms over: 1

(a) warm oceans **(b)** cold deserts **(c)** poles **(d)** mountains

10. A sharp knife cuts better than a blunt one because: 1

(a) it is heavier
(b) its edge has a larger area
(c) its edge has a smaller area, so pressure is greater
(d) it is bigger

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Define pressure and write its formula and SI unit. 2

12. What is atmospheric pressure? 2

13. What is the 'eye' of a cyclone? 2

14. Why does a nail have a pointed end? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A box of weight 600 N rests on a floor with a base of area 0.5 square metre. Find the pressure. 3

16. Why does a balloon expand when more air is blown in? 3

17. Why do we feel the wind more near the sea during the day? 3

OR

Explain why it is easier to cut with a sharp knife than a blunt one. 3

18. Explain the formation of a cyclone. 3

OR

Explain why a gas fills the whole container and exerts pressure on its walls. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain, with examples, how pressure depends on force and area. Calculate the pressure of a 50 N force over 0.25 square metre and over 0.5 square metre. 5

OR

Explain the working of a barometer and the variation of atmospheric pressure with altitude and weather. 5

20. Explain how pressure in liquids depends on depth and how it acts in all directions. Describe an activity. 5

OR

List safety precautions to take before, during and after a cyclone. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND